

CONTROL SYSTEMS LABORATORY

Experiment No:	7
Experiment Name:	TRANSIENT RESPONSE ANALYSIS
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1. OBJECTIVES

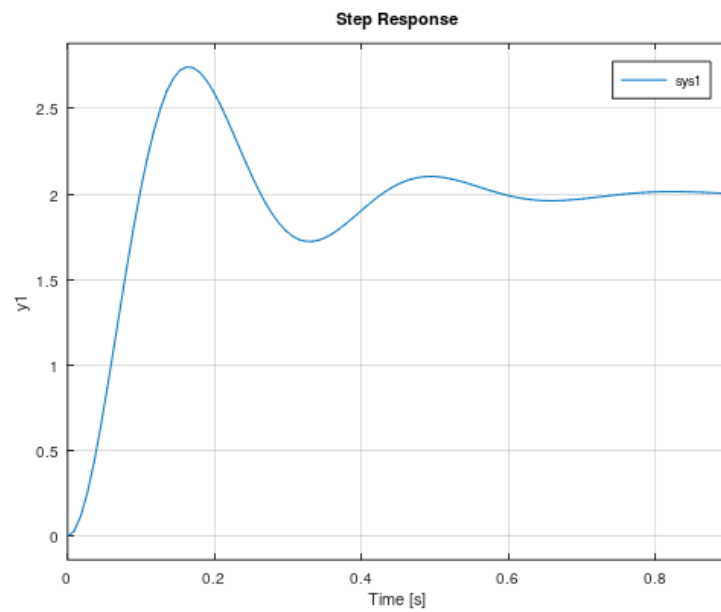
The main purpose of software experiments is to introduce the computer-aided control system analysis and design. In this experiment, you will practice about transient response analysis by using Octave.

2. EXPERIMENTAL RESULTS

A-

```
Wn1 =  
  
    20  
    20  
  
zetal =  
  
    0.3000  
    0.3000  
  
poles1 =  
  
    -6.000 + 19.079i  
    -6.000 - 19.079i  
  
>> |
```

```
>> Wn11 = sqrt(400)  
zetal1 = 12/(2*Wn11)  
poles11 = [-(zetal1*Wn11)+Wn11*sqrt((zetal1^2)-1) -(zetal1*Wn11)-Wn11*sqrt((zetal1^2)-1)]  
[z1,p1,k1]=tf2zp(sys1)  
Wn11 = 20  
zetal1 = 0.3000  
poles11 =  
  
    -6.000 + 19.079i    -6.000 - 19.079i  
  
z1 = [] (0x1)  
p1 =  
  
    -6.000 + 19.079i  
    -6.000 - 19.079i  
  
k1 = 800  
>> |
```



First system is underdamped.

```
>> [Wn2, zeta2, poles2] = damp(sys2)
Wn2 = sqrt(900)
zeta22 = 90/(2*Wn22)
poles22 = [-(zeta22*Wn22)+Wn22*sqrt((zeta22^2)-1) -(zeta22*Wn22)-Wn22*sqrt((zeta22^2)-1)]
[z2,p2,k2]=tf2zp(sys2)
Wn2 =

    11.459
    78.541

zeta2 =

     1
     1

poles2 =

    -11.459
    -78.541

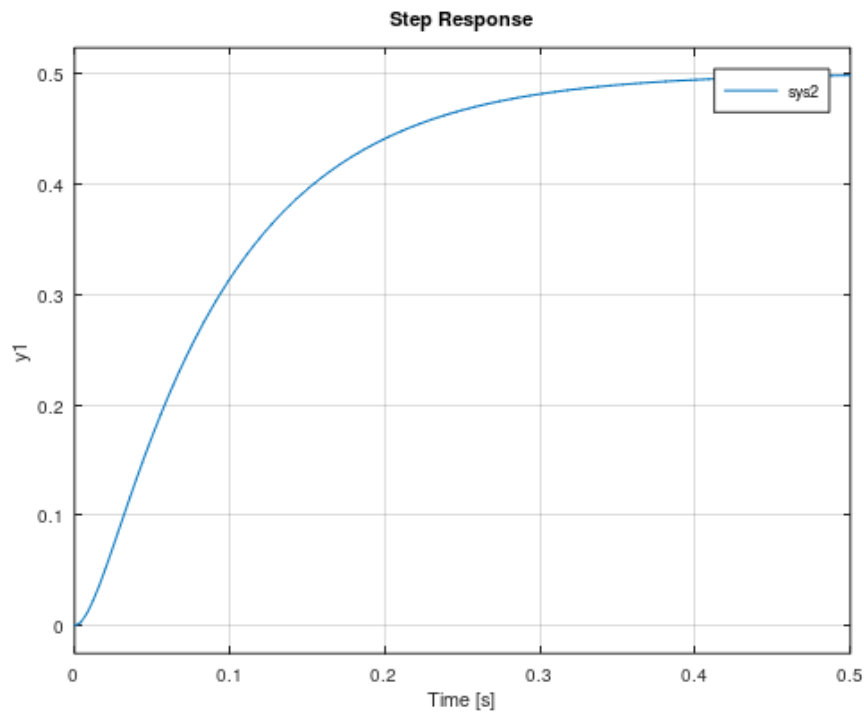
Wn2 = 30
zeta22 = 1.5000
poles22 =

    -11.459    -78.541

z2 = [] (0x1)
p2 =

    -78.541
    -11.459

k2 = 450
>> |
```



This system is overdamped.

```
>> [Wn3, zeta3, poles3] = damp(sys3)
Wn3 = sqrt(225)
zeta33 = 30/(2*Wn33)
poles33 = [-(zeta33*Wn33)+Wn33*sqrt((zeta33^2)-1) -(zeta33*Wn33)-Wn33*sqrt((zeta33^2)-1)]
[z3,p3,k3]=tf2zp(sys3)
figure(3)
step(sys3)
Wn3 =
    15
    15

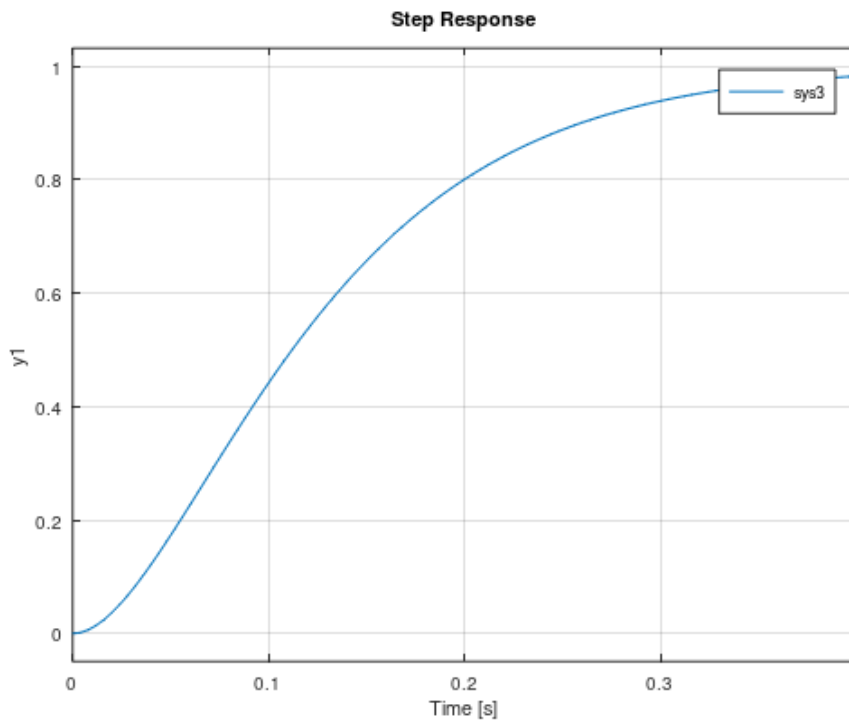
zeta3 =
     1
     1

poles3 =
   -15
   -15

Wn33 = 15
zeta33 = 1
poles33 =
   -15   -15

z3 = [] (0x1)
p3 =
   -15
   -15

k3 = 225
>> |
```



This system is critically damped.

```
>> [Wn4, zeta4, poles4] = damp(sys4)
Wn4 = sqrt(625)
zeta4 = 0/(2*Wn4)
poles4 = [-(zeta4*Wn4)+Wn4*sqrt((zeta4^2)-1) -(zeta4*Wn4)-Wn4*sqrt((zeta4^2)-1)]
[z4,p4,k4]=tf2zp(sys4)
figure(4)
step(sys4)
Wn4 =

    25
    25

zeta4 =

   -6.1230e-17
   -6.1230e-17

poles4 =

    0 + 25i
    0 - 25i

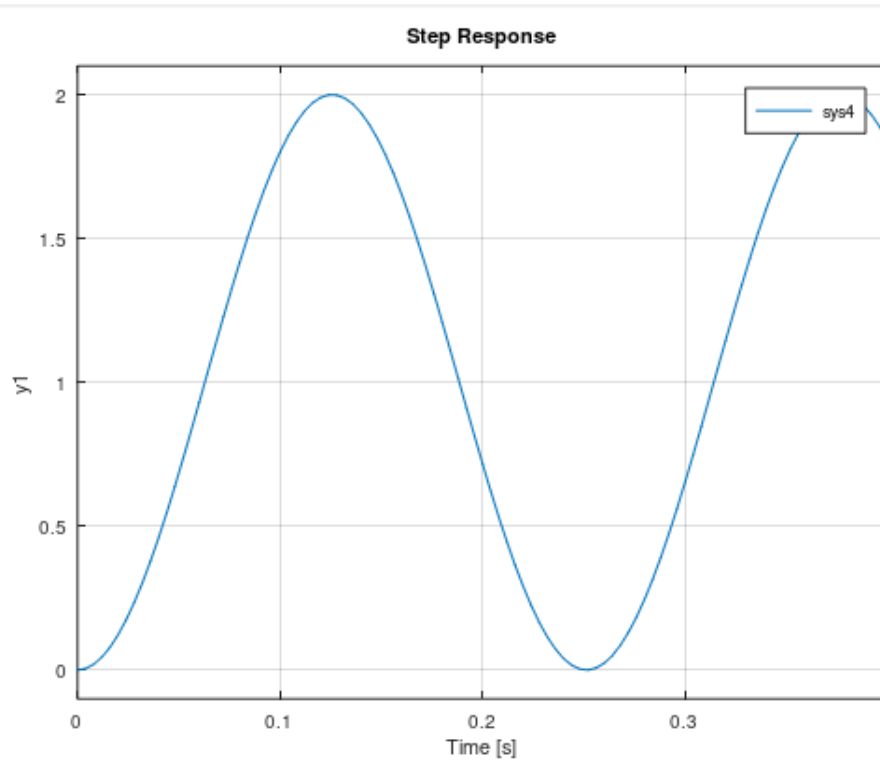
Wn4 = 25
zeta4 = 0
poles4 =

    0 + 25i    0 - 25i

z4 = [] (0x1)
p4 =

    0 + 25i
    0 - 25i

k4 = 625
>> |
```



This system is underdamped

B-

Transfer function 'sys' from input 'u1' to output ...

$$y1: \frac{12}{s^2 + 8s + 12}$$

Continuous-time model.

Transfer function 'sys' from input 'u1' to output ...

$$y1: \frac{16}{s^2 + 8s + 16}$$

Continuous-time model.

Transfer function 'sys' from input 'u1' to output ...

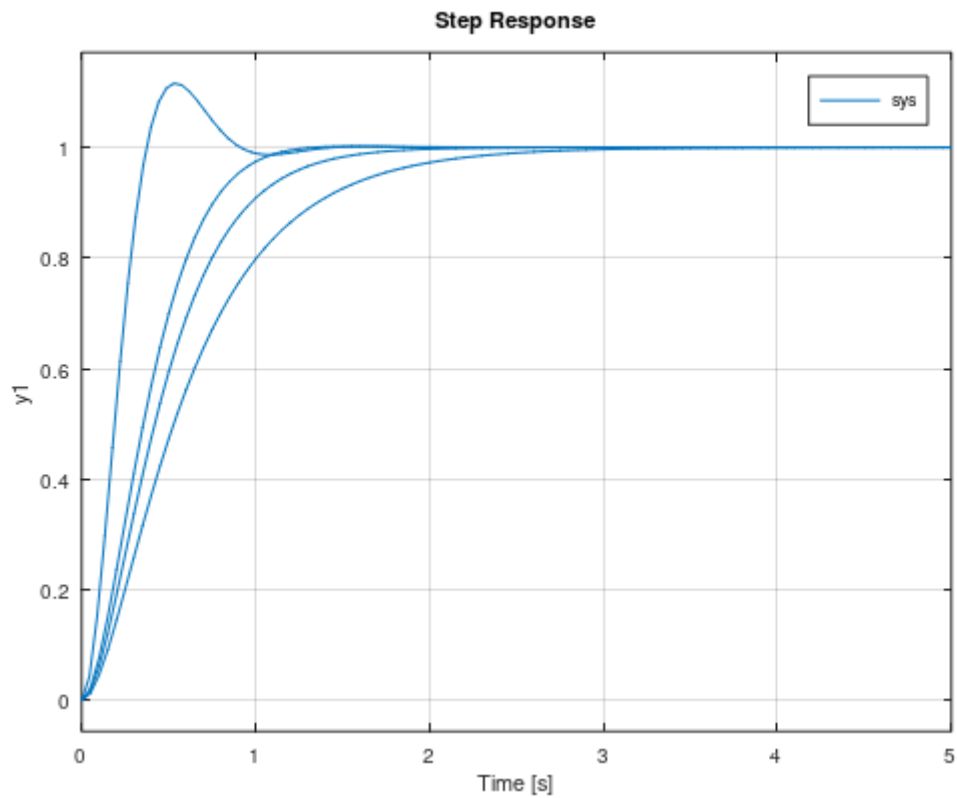
$$y1: \frac{20}{s^2 + 8s + 20}$$

Continuous-time model.

Transfer function 'sys' from input 'u1' to output ...

$$y1: \frac{50}{s^2 + 8s + 50}$$

Continuous-time model.



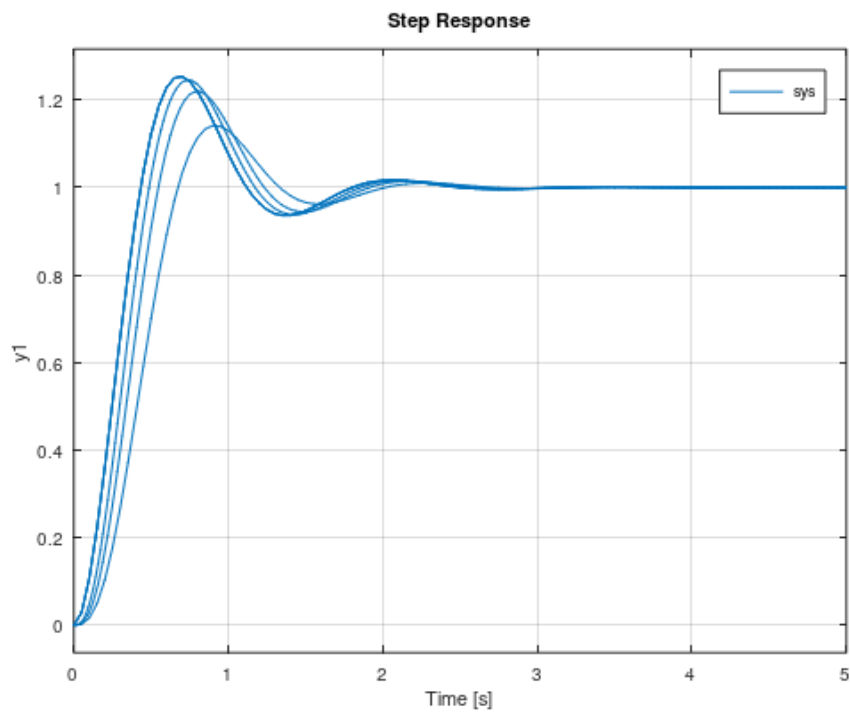
The first system is overdamped.

The second system is critically damped.

The third system is underdamped.

The fourth system is underdamped.

C-



Transfer function 'sys_new' from input 'u1' to output ...

$$y1: \frac{125}{s^3 + 9 s^2 + 45 s + 125}$$

Continuous-time model.

Transfer function 'sys_new' from input 'u1' to output ...

$$y1: \frac{250}{s^3 + 14 s^2 + 65 s + 250}$$

Continuous-time model.

Transfer function 'sys_new' from input 'u1' to output ...

$$y1: \frac{500}{s^3 + 24 s^2 + 105 s + 500}$$

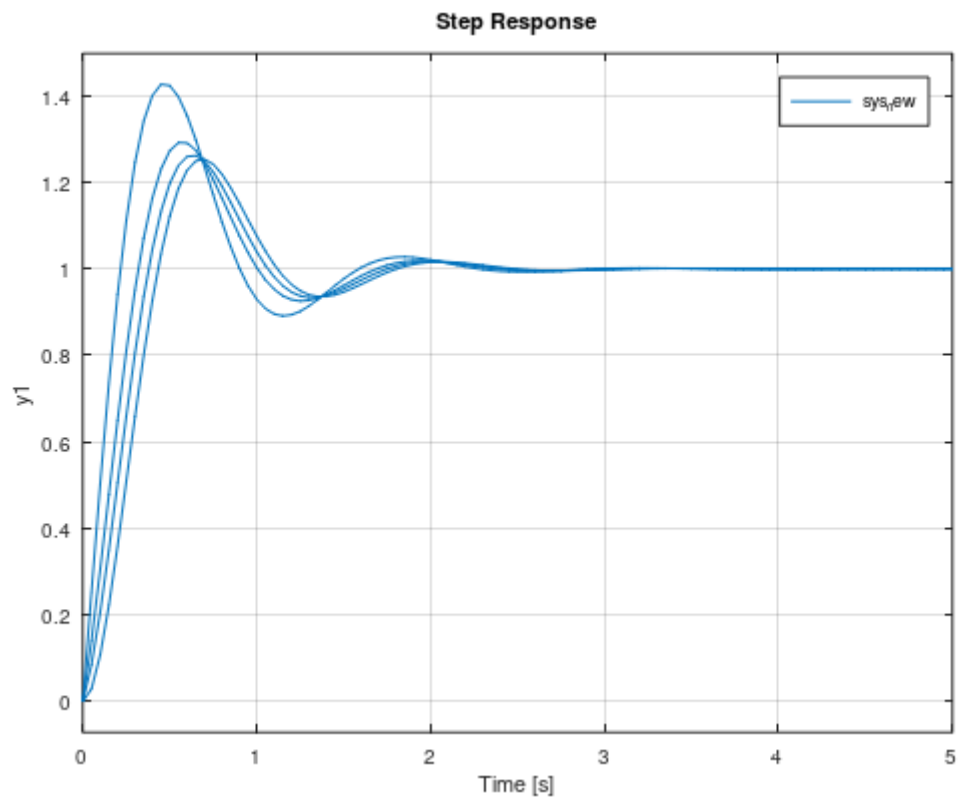
Continuous-time model.

>> |

Transient response specifications for given γ ($-5, 10, 20$) values are

γ values	pole locations	t_s (sec)	M_p (%)
-	Poles: $-2 \pm 4.582i$	1.68	25.4
5	Poles: $-2 \pm 4.582i, -5$	1.83	14.1
10	Poles: $-2 \pm 4.582i, -10$	1.78	21.9
20	Poles: $-2 \pm 4.582i, -20$	1.73	24.5

D-



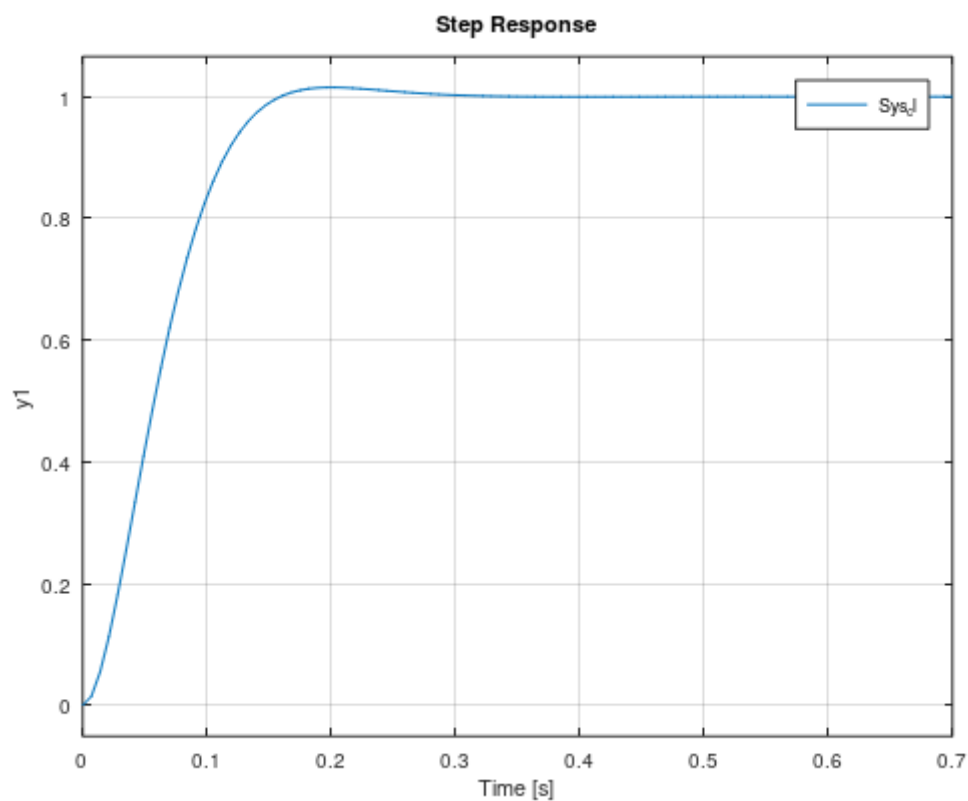
E-

```
zeta_2 = 0.7156
>> zeta = 0.8;
wn = pi/(0.2*sqrt(1-zeta^2));
>> K1_value = wn^2;
K2_value = ((2*zeta*wn)-1)/K1_value;
>> G = tf([K1_value],[1 1 0]);
H = tf([K2_value 1],[1]);
Sys_cl = feedback(G,H)
step(Sys_cl)

Transfer function 'Sys_cl' from input 'u1' to output ...

          685.4
y1:  -----
      s^2 + 41.89 s + 685.4

Continuous-time model.
>>
>>
>> |
```



Peak time : 0.2

Overshoot : %1.52

3. CONCLUSION & DISCUSSION

In conclusion, we tested how the system's overshoot and settling time are affected.

We identified the types of systems. We have seen that the farther the zero value gets from the origin, the less its effect. We have seen that the closer we get, the greater the effect. We also saw that the added pole value has less effect as it moves away from the dominant one. We saw that it increased as we got closer. If the added root value moves away from the dominant by 5 times, its effect is neglected.

REFERENCES

Write your reference list in this section by using numbering with square brackets - []. You should follow IEEE Citation StyleGuide for referencing format.

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021